

E-Commerce SQL Project

Project Title

E-Commerce Sales Analysis Using MySQL

Project Overview

This project focuses on developing an **E-Commerce Management and Sales Analysis System using MySQL**. The database manages **categories, products, customers, and orders** and uses SQL queries to analyze product pricing, inventory, customers, orders, and sales.

The project demonstrates practical SQL concepts including **filtering, aggregation, grouping, joins, subqueries, conditional logic, sorting, updating, and deleting records**.

Objectives

- Create an e-commerce relational database.
- Manage product and category information.
- Store customer details.
- Manage customer orders.
- Establish relationships using primary and foreign keys.
- Analyze product prices and stock.
- Analyze customers by city.
- Calculate total sales.
- Analyze product-wise sales.
- Generate business insights using SQL.

Database

Database Name: ecommerce

Tables:

- Categories
- Products
- Customers
- Orders

Dataset Information

Categories

- Category ID
- Category Name

Categories include:

- Electronics
- Clothing
- Shoes
- Home
- Books

Products

- Product ID
- Product Name
- Price
- Stock
- Category ID

Products include:

- Laptop
- Mobile Phone
- T-Shirt
- Jeans
- Running Shoes

- Sofa
- Table Lamp
- SQL Book
- Python Book
- Smart Watch

Customers

- Customer ID
- Customer Name
- City
- Email

Orders

- Order ID
- Customer ID
- Product ID
- Quantity
- Order Date
- Status

Order statuses include:

- Delivered
- Pending
- Shipped
- Cancelled

SQL Concepts Used

- Database Creation
- Table Creation
- Primary Keys
- Foreign Keys
- Constraints
- SELECT
- INSERT
- UPDATE
- DELETE
- WHERE
- BETWEEN
- IN
- LIKE
- OR
- GROUP BY

- HAVING
- ORDER BY
- INNER JOIN
- Subqueries
- CASE Statements
- Aggregate Functions
 - COUNT ()
 - SUM ()
 - AVG ()
 - MAX ()
 - MIN ()

Analysis Performed

1. Product Analysis

Analyzed:

- All products
- Products costing more than 5,000
- Products between 1,000 and 10,000
- Products from selected categories
- Products starting with a specific letter
- Products with stock below 20
- Maximum product price
- Minimum product price
- Average product price
- Total product stock

2. Customer Analysis

Analyzed:

- Customers from Pune
- Customers from Mumbai
- Customers from Pune or Mumbai
- Customer order details

3. Category Analysis

Performed:

- Product count by category
- Average price by category
- Categories having more than two products

4. Order Analysis

Analyzed:

- Customer and order details
- Complete order information
- Order status
- Product quantity
- Product price
- Order amount

5. Sales Analysis

Calculated:

Order Amount

`Quantity × Product Price`

Total Sales

`SUM(Quantity × Product Price)`

Also performed **product-wise sales analysis**.

6. Subquery Analysis

Used subqueries to identify:

- Products costing more than average price
- Most expensive product

7. Price Categorization

Used `CASE` statements to classify products as:

Price	Category
₹30,000+	Expensive
₹4,000-₹29,999	Medium
Below ₹4,000	Cheap

8. Data Modification

Performed:

- Laptop price update
- Deletion of cancelled orders

Key Business Insights

- Identified products with high and low prices.
- Identified products with low stock levels.
- Calculated average product pricing.
- Analyzed product distribution across categories.
- Compared average prices by category.
- Identified customers by location.
- Calculated individual order amounts.
- Calculated total sales.
- Identified product-wise sales.
- Found products priced above the average.
- Identified the most expensive product.
- Categorized products according to price levels.

SQL Queries Included

The project contains queries for:

- Database and table creation
- Data insertion
- Product filtering
- Customer filtering
- Product statistics
- Category analysis
- Customer analysis
- Order analysis
- Multiple-table joins
- Sales calculations
- Product-wise sales
- Subqueries
- Price categorization
- Sorting
- Updating records
- Deleting cancelled orders

Skills Demonstrated

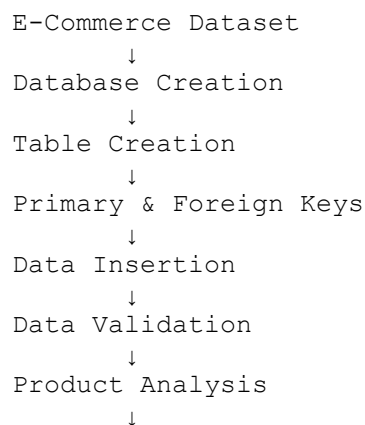
- MySQL
- SQL
- Relational Database Design
- Primary & Foreign Keys
- Data Analysis
- Exploratory Data Analysis
- Aggregate Functions
- Joins
- Subqueries
- GROUP BY
- HAVING
- CASE
- Data Interpretation
- Business Problem Solving

Business Value

The project helps an e-commerce business answer questions such as:

- Which products are expensive?
- Which products have low stock?
- Which categories contain more products?
- What is the average product price?
- How much total sales were generated?
- Which products contribute to sales?
- Which customers are from specific cities?
- Which products are priced above average?
- What is the most expensive product?
- How can products be classified by price?

End-to-End Project Flow



Customer Analysis
↓
Category Analysis
↓
Order Analysis
↓
Sales Analysis
↓
JOIN & Subquery Analysis
↓
CASE Categorization
↓
Business Insights

Conclusion

This **E-Commerce Sales Analysis Using MySQL** project demonstrates how SQL can be used to manage relational business data and generate meaningful insights. The project covers **products, categories, customers, orders, inventory, pricing, and sales** while applying important SQL concepts such as **filtering, aggregation, grouping, joins, subqueries, conditional logic, updating, and deleting records**.

The project provides practical experience relevant to **Data Analyst, Business Analyst, SQL Developer, and Database-related roles**.

Tools Used

- MySQL
- SQL
- MySQL Workbench
- Relational Database